








Advanced Time Series Analysis and Forecasting

Chapter 16: Explosive roots and bubbles



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Acronyms used in this material (1/2)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **BSADF**: Backward Supremum Augmented Dickey–Fuller (statistic)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **CLT**: Central Limit Theorem
- **CPI**: Consumer Price Index
- **CUSUM**: CUmulative SUM
- **EODHD**: EOD Historical Data (market-data provider)
- **FCLT**: Functional Central Limit Theorem
- **FRED**: Federal Reserve Economic Data
- **FW**: Family-Wise (error control)
- **FWER**: Family-Wise Error Rate

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GSADF**: Generalised Supremum Augmented Dickey–Fuller (test)
- **HICP**: Harmonised Index of Consumer Prices
- **LLM**: Large Language Model
- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **MC**: Monte Carlo
- **MFM**: Modelling Financial Markets
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips, Shi and Yu (2015)
- **PWY**: Phillips, Wu and Yu (2011)
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **ROC**: Receiver Operating Characteristic (curve)
- **SADF**: Supremum Augmented Dickey–Fuller (test)
- **TSA**: Time Series Analysis
- **US**: United States
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

The question of the chapter and the route

- **Question**: when does a price series contain an explosive component, how precisely can we date it, and how much is an alarm worth once false alarms and base rates are counted?
  - three layers: economic theory of rational bubbles, asymptotic theory of explosive autoregressions, statistical evaluation of detectors
- **Route**
  - rational bubbles: present value, Blanchard–Watson, the Diba–Grossman critique, Evans bubbles
  - explosive autoregressions: Cauchy limits, mildly explosive roots; SADF, GSADF, BSADF and their size
  - date-stamping, real-time monitoring, evaluation of early warnings; fundamentals against prices (rents)
  - LPPLS as a competing detector; applications: dot-com, Shanghai 2015, Bitcoin, BET 2007, housing
- Prerequisites and continuation
  - TSA, Chapter 13 (slide Prior knowledge from TSA and new material); Chapter 2: unit roots, structural breaks
  - MFM, Chapter 17 applies the tests to markets

Self-study guide

- This chapter is for **self-study**: there is no seminar; every section ends with a recap
  - redo each worked derivation on paper before reading the next slide
  - after each chart, write your own reading of it, then compare it with the interpretation slide
- Run the lecture notebook: it reproduces every chart and number with smaller Monte Carlo sizes
  - change one design choice (the minimum window, the lag order, the critical value, the crash threshold) and record how the conclusion changes
- The self-assessment at the end has answers; the quiz of this chapter on the course site is graded
- Prerequisites: Dickey–Fuller asymptotics (Hamilton, Chapter 17), the functional central limit theorem, bootstrap for dependent data (Chapter 0)

Learning outcomes

- Derive the general solution of the present-value model and state the restrictions of Diba and Grossman on rational bubbles
- Derive the Cauchy limit of the OLS estimator under an explosive root and explain why the mildly explosive limit is invariant to the error law
- Compute SADF, GSADF and BSADF with Monte Carlo and wild-bootstrap critical values, and control false alarms across dates
- Measure date-stamping delays and evaluate early-warning signals with hit rates, false-alarm rates, precision and ROC curves
- Separate explosive prices from explosive fundamentals, and compare PSY with LPPLS confidence indicators on real episodes

Reading, data and tools

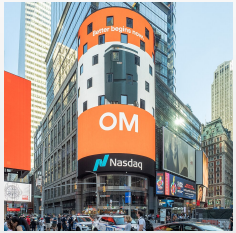
- Theory: Blanchard and Watson (1982); Diba and Grossman (1988a); Evans (1991); Phillips and Magdalinos (2007); Phillips, Shi and Yu (2015a); Phillips, Shi and Yu (2015b)
  - Inference and monitoring: Phillips and Shi (2020); Harvey et al. (2016); Homm and Breitung (2012); Astill et al. (2018); survey: Gürkaynak (2008)
  - LPPLS: Johansen, Ledoit and Sornette (2000); Filimonov and Sornette (2013); Filimonov, Demos and Sornette (2017); Shu and Zhu (2020); critique: Feigenbaum (2001); Geraskin and Fantazzini (2013)
- Python Quantlets of this chapter: Quantlets/Ch_16
  - every window regression written out in `numpy` (cumulated cross-products); the R package `exuber` Vasilopoulos, Pavlidis and Martínez-García (2022) implements the same statistics
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Sample
S&P 500, Nasdaq 100, Shanghai Composite, BET, Bitcoin	EODHD daily closes (weekly: Friday close)	1990 – 2026
US house prices, rents, consumer prices	FRED: S&P CoreLogic Case–Shiller national index, CPI rent of primary residence, CPI (seasonally adjusted)	1987 – July 2026
Romanian house prices, rents, consumer prices	Eurostat: house price index; HICP actual rentals and all items	2009 – March 2026
All price series of the course data	EODHD, monthly closes (screen of the AI mini-case)	1990 – 2026

- Tests run on log prices or log ratios; weekly data for stock indices and Bitcoin, monthly or quarterly data for housing

Prior knowledge from TSA and new material



Nasdaq MarketSite, Times Square, 2021

Photo: ajay_suresh (2021); CC BY 2.0; Wikimedia Commons

- Known (TSA, Chapter 13)
  - manias in history; the right-tailed ADF, SADF, GSADF and BSADF with Monte Carlo critical values
  - the LPPL equation, the two-step calibration, the confidence indicator
- New: the research layer
  - why rational bubbles must explode and why they are hard to see
  - the limit theory behind the tests, their size under changing volatility and across many dates
  - how early, how often and how wrongly the detectors raise alarms
- Case studies, on our data
  - Phillips, Wu and Yu (2011); Phillips, Shi and Yu (2015a); Phillips and Yu (2011)
  - Homm and Breitung (2012); Harvey et al. (2016); Sornette et al. (2015)

The present-value model and its bubble solutions (1/2)

- **No-arbitrage** with a constant required return $r > 0$: today's price is the discounted expected value of tomorrow's price plus dividend

$$P_t = (1+r)^{-1}\mathbb{E}_t[P_{t+1} + D_{t+1}]$$

  - P_t: price; D_t: dividend (or rent); $\mathbb{E}_t$: expectation given the information available at t

- **Forward substitution** k times: the price is the discounted dividends of the next k periods plus the discounted price at $t+k$

$$P_t = \sum_{i=1}^{k}(1+r)^{-i}\mathbb{E}_t D_{t+i} + (1+r)^{-k}\mathbb{E}_t P_{t+k}$$

- If the **transversality condition** $\lim_{k\to\infty}(1+r)^{-k}\mathbb{E}_t P_{t+k} = 0$ holds, the price equals the **fundamental value**

$$P_t = F_t := \sum_{i\geq 1}(1+r)^{-i}\mathbb{E}_t D_{t+i}$$

The present-value model and its bubble solutions (2/2)

- **General solution**: the fundamental plus a **bubble** B_t whose expected value grows at the rate r

$$P_t = F_t + B_t, \qquad \mathbb{E}_t B_{t+1} = (1+r)B_t, \qquad \mathbb{E}_t B_{t+k} = (1+r)^k B_t$$

  - B_t is a submartingale with an explosive mean: it satisfies the no-arbitrage equation but violates transversality
- **Econometric content**: B_t follows $(1-(1+r)L)B_t = z_t$, an AR(1) with coefficient $1+r>1$
  - L: the lag operator ($LB_t = B_{t-1}$); z_t: a martingale difference ($\mathbb{E}_{t-1} z_t = 0$); an **explosive autoregressive root**
- General equilibrium limits Tirole (1985); Santos and Woodford (1997)
  - with infinitely lived agents bubbles are ruled out in most economies
  - with overlapping generations they can exist if growth exceeds r

Worked example: the fundamental value with random-walk dividends

- Let dividends be a random walk with drift, $D_t = \mu + D_{t-1} + \varepsilon_t$: then $\mathbb{E}_t D_{t+i} = D_t + i\mu$
  - μ: the expected change per period (drift); ε_t: a zero-mean shock
- Summing the discounted expected dividends

$$F_t = D_t \sum_{i \geq 1} (1+r)^{-i} + \mu \sum_{i \geq 1} i(1+r)^{-i} = \frac{D_t}{r} + \frac{\mu(1+r)}{r^2}$$

  - using $\sum_{i \geq 1} x^i = x/(1-x)$ and $\sum_{i \geq 1} i x^i = x/(1-x)^2$ with $x = (1+r)^{-1}$
- Implications: F_t is I(1) like D_t; $P_t - D_t/r$ is stationary without a bubble: price and dividend are **cointegrated** with vector $(1, -1/r)$ Campbell and Shiller (1987)
  - with a bubble, $P_t - D_t/r - \mu(1+r)/r^2 = B_t$ inherits the explosive root: no difference $\Delta^d P_t$ is stationary Diba and Grossman (1988b)
- Check on paper: $r = 0.05$, $\mu = 0.04$, $D_t = 2$ give $F_t = 40 + 16.8 = 56.8$; the price-dividend ratio 28.4 is constant only if $\mu = 0$

The Blanchard–Watson stochastic bubble



Olivier Blanchard, IMF, 2008

Photo: Eugene Salazar, IMF (2008); public domain; Wikimedia Commons

- Blanchard and Watson (1982): each period the bubble survives with probability π and bursts with probability $1-\pi$

$$B_{t+1} = \begin{cases} \dfrac{1+r}{\pi} B_t + \varepsilon_{t+1} & \text{with probability } \pi \\ \varepsilon_{t+1} & \text{with probability } 1-\pi \end{cases}$$

  - ε_{t+1}: a zero-mean shock; $\pi \in (0,1)$: the survival probability
- Check: $\mathbb{E}_t B_{t+1} = \pi \cdot \frac{1+r}{\pi} B_t = (1+r) B_t$: a valid rational bubble
- Survival time is geometric: expected duration $1/(1-\pi)$; $\pi = 0.98$ gives 50 periods
- While it survives it grows at $(1+r)/\pi - 1 > r$: the extra return pays for the crash risk
- After a burst it restarts from noise, which can be negative: the next slide shows why that is a problem

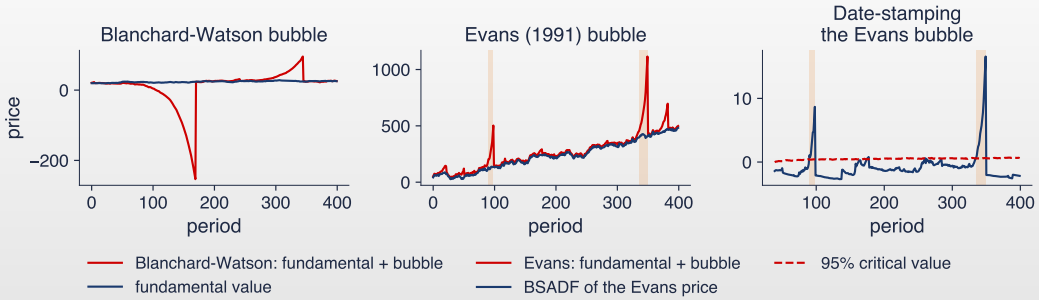
The Diba–Grossman critique

- Write $B_{t+1} = (1+r)B_t + z_{t+1}$ with $\mathbb{E}_t z_{t+1} = 0$ Diba and Grossman (1988a)
- **No negative bubbles**: if $B_t < 0$, then $\mathbb{E}_t B_{t+k} = (1+r)^k B_t \to -\infty$, so the expected price eventually becomes negative; free disposal ($P \geq 0$) rules this out
- **No birth**: if $B_t = 0$, then $\mathbb{E}_t B_{t+1} = 0$ and $B_{t+1} \geq 0$, hence $B_{t+1} = 0$ almost surely; a bubble that exists today existed on the first trading day
  - the Blanchard–Watson restart from noise violates both points; a bubble can still burst, but not restart
- **Test** Diba and Grossman (1988b): no bubble $\Rightarrow$ P_t and D_t cointegrated and ΔP_t stationary; on annual US data they found no explosive component
  - unobserved fundamentals or a time-varying r make the test a joint test of the bubble and of the pricing model Gürkaynak (2008)

Evans: periodically collapsing bubbles

- Parameters $0 < \delta < (1+r)\alpha$, $u_{t+1} > 0$ i.i.d. with $\mathbb{E}u = 1$, $\theta_{t+1} \sim$ Bernoulli(π) Evans (1991)
  - $B_{t+1} = (1+r)B_t u_{t+1}$ if $B_t \leq \alpha$ (slow growth phase)
  - $B_{t+1} = \left[\delta + \pi^{-1}(1+r)\theta_{t+1}\left(B_t - \delta/(1+r)\right)\right]u_{t+1}$ if $B_t > \alpha$ (fast phase that collapses to δ)
- Exercise: show $\mathbb{E}_t B_{t+1} = (1+r)B_t$ in both regimes (use $\mathbb{E}\theta = \pi$); the bubble is positive and never dies, consistent with Diba–Grossman
- Evans' point: on the whole sample, frequent collapses make P_t look like an I(1) or even stationary process; unit-root and cointegration tests have almost no power
  - the full-sample OLS slope pairs the largest falls with the highest lagged levels: collapses pull $\hat{\delta}$ down
- The remedy of Phillips, Wu and Yu (2011) and Phillips, Shi and Yu (2015a): estimate on sub-samples (windows) and take the supremum, so that some windows contain the expansion but not the collapse

Simulated rational bubbles and their detection



- Left: Blanchard–Watson, $r = 0.02$, $\pi = 0.98$; middle: Evans bubble times 20 on the fundamental of the worked example ($r = 0.05$, $\alpha = 1$, $\delta = 0.5$, $\pi = 0.85$); right: BSADF of the Evans price against its pointwise 95% critical value (2 000 Monte Carlo paths)
- Whole-sample ADF -2.81 (95% critical value -0.07); SADF 6.80 (1.45); GSADF 16.58 (2.18); 2 dated episodes

ATS_ch16_rational_bubbles

Interpreting the simulated bubbles

- The Blanchard–Watson path restarts below zero after a burst and then explodes downwards: exactly the negative bubble that Diba and Grossman exclude
- The Evans price looks like its fundamental most of the time; the whole-sample ADF (-2.81) is even below its 95% critical value: Evans' pitfall in one number
- The window statistics see the large expansions: GSADF 16.58 against 2.18; the BSADF sequence crosses its critical value only in the long runs
- The minimum-duration rule (6 consecutive exceedances) keeps 2 episodes and drops the short bursts: power is traded for fewer false alarms

Recap: rational bubbles

- Price = fundamental + bubble, with $\mathbb{E}_t B_{t+1} = (1 + r)B_t$: a bubble is an explosive autoregressive component
- Diba–Grossman: bubbles cannot be negative and cannot start later; they can only grow or burst
- Evans: periodic collapses hide the bubble from whole-sample tests; recursive window tests restore the power
- Every bubble test is a joint test with the model of the fundamental (a constant r, observed dividends or rents)

Three regimes of the AR(1) estimator (1/2)

Root	**Rate and limit of $\hat{\rho} - \rho$**	**Invariance**
$\lvert\rho\rvert < 1$	$\sqrt{n}(\hat{\rho} - \rho) \Rightarrow N(0, 1 - \rho^2)$	yes (CLT)
$\rho = 1$	$n(\hat{\rho} - 1) \Rightarrow \int_0^1 W\, dW / \int_0^1 W^2$	yes (FCLT)
$\rho = 1 + c/k_n$, $k_n \to \infty$, $k_n = o(n)$	$\frac{k_n \rho_n^n}{2c}(\hat{\rho} - \rho_n) \Rightarrow \mathcal{C}$	yes Phillips and Magdalinos (2007)
$\lvert\rho\rvert > 1$ fixed	$\frac{\rho^n}{\rho^2 - 1}(\hat{\rho} - \rho) \Rightarrow \mathcal{C}$	only Gaussian errors White (1958); Anderson (1959)

- Model: $y_t = \rho y_{t-1} + u_t$, $y_0 = 0$, u_t i.i.d. $(0, \sigma^2)$; $\hat{\rho}$: OLS without intercept
  - n: sample size; W: a standard Brownian motion; $\mathcal{C}$: the standard Cauchy law; $\Rightarrow$: convergence in distribution
  - CLT/FCLT: (functional) central limit theorem

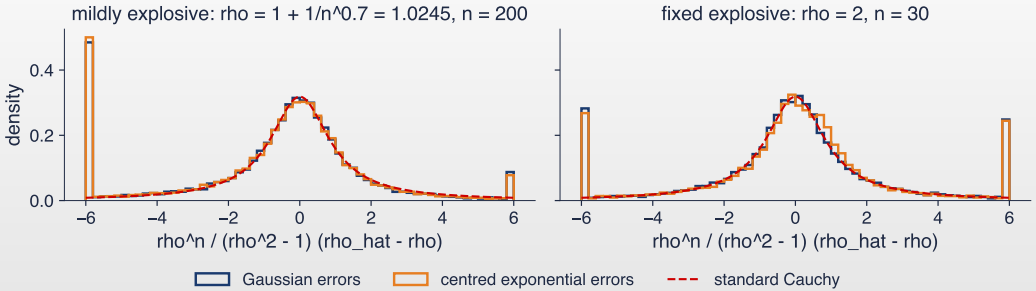
Three regimes of the AR(1) estimator (2/2)

- Mildly explosive root: $\rho_n = 1 + c/k_n$
  - $c > 0$, and $k_n \to \infty$ more slowly than n: $k_n = o(n)$, i.e. $k_n/n \to 0$
  - the Cauchy limit is derived in the Appendix
- The rate grows from $\sqrt{n}$ to n to the exponential ρ^n
  - the further from unity, the faster we learn ρ
  - unit roots: Chapter 2 and Hamilton (1994), Chapter 17
- Mildly explosive roots are the realistic case for bubbles
  - faster than any local-to-unity root $1 + c/n$
  - slower than any fixed explosive root
  - their limit theory is invariant to the error law, unlike the fixed explosive case

Worked derivation of the Cauchy limit

- $\hat{\rho} - \rho = \sum_t y_{t-1} u_t / \sum_t y_{t-1}^2$ and $y_t = \sum_{j \le t} \rho^{t-j} u_j$
- Define $Y_n = \sum_{j=1}^{n} \rho^{-j} u_j$ (dominated by the **first** shocks) and $X_n = \sum_{j=1}^{n} \rho^{-(n-j)-1} u_j$ (dominated by the **last** shocks)
  - then $\rho^{-n} y_n \approx Y_n$, $\rho^{-2n} \sum_t y_{t-1}^2 \approx Y_n^2/(\rho^2 - 1)$ and $\rho^{-n} \sum_t y_{t-1} u_t \approx X_n Y_n$
- Hence $\dfrac{\rho^n}{\rho^2 - 1}(\hat{\rho} - \rho) \approx \dfrac{X_n}{Y_n}$
  - X_n and Y_n use disjoint blocks of shocks asymptotically: independent, with equal variances $\sigma^2/(\rho^2 - 1)$
  - **fixed** ρ: each sum is dominated by a few shocks; it is Gaussian only if the u_j are Gaussian, so the ratio is Cauchy only then Anderson (1959)
  - **mildly explosive** ρ_n: about $k_n \to \infty$ shocks enter each sum with comparable weights, a CLT applies, and $X_n/Y_n \Rightarrow \mathcal{C}$ for any i.i.d. errors with finite variance Phillips and Magdalinos (2007)
- Pivotal inference: the 95% interval $\hat{\rho} \pm (\hat{\rho}^2 - 1)\hat{\rho}^{-n} \tan(0.475\pi)$, with $\tan(0.475\pi) \approx 12.71$, replaces ρ by $\hat{\rho}$ in the norming factor

Cauchy limits in finite samples



- 20 000 samples; normalised error $\rho^n(\hat{\rho} - \rho)/(\rho^2 - 1)$ with Gaussian and centred exponential errors (skewed), against the Cauchy density
- Kolmogorov–Smirnov distance to Cauchy: mild, $n = 200$: 0.060 and 0.062; $n = 2000$ ($\rho = 1.0049$): 0.011 ($p = 0.620$) and 0.007 ($p = 0.945$); fixed $\rho = 2$: 0.007 ($p = 0.263$) and 0.025 ($p < 0.001$)

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Interpreting the Cauchy limits

- Fixed root: with Gaussian errors the Cauchy law fits ($p = 0.263$); with skewed errors it is rejected ($p < 0.001$): no invariance principle, as Anderson showed
- Mildly explosive root: at $n = 2000$ both error laws give the Cauchy law; at $n = 200$ ($k_n = 40.8$) the approximation is still rough for both: the CLT inside X_n and Y_n needs many effective shocks
- Coverage of the 95% Cauchy interval: 97.0% and 97.2% ($n = 200$), 95.0% and 94.7% ($n = 2000$): conservative in small samples, accurate in large ones
- For bubble tests this is good news: the explosive alternative is learned at an exponential rate, so the window statistics diverge fast once a window sits inside a bubble

Limit theory and the consistency of bubble tests

- Under a mildly explosive alternative the ADF t-statistic diverges to $+\infty$: right-tailed tests are **consistent** Phillips, Shi and Yu (2015b)
- Levels against logs: a rational bubble is explosive in the **level**; in logs a pure bubble path $\ln B_0 + t \ln(1+r)$ is a linear trend
  - $\ln(F_t + B_t)$ **accelerates** while the bubble share rises: log-price tests target accelerating log prices, which is what PWY and PSY test in practice
- Specification matters: the intercept, a weak drift $dT^{-\eta}$ ($\eta > 1/2$) in the null, and the lag order change the null limit and the size Phillips, Shi and Yu (2014)
- Collapses: a sudden fall is a mildly **integrated** or stationary phase after the explosive one; Phillips and Shi (2018) date the implosion with a reverse regression

Recap: explosive autoregressions

- Explosive roots are learned at rate ρ^n; the limit is Cauchy, a ratio of two independent normals built from the first and the last shocks
- Fixed explosive roots have no invariance principle; mildly explosive roots do, which makes them the working model for bubbles
- The Cauchy interval for ρ is pivotal but needs large samples; right-tailed tests are consistent against mildly explosive alternatives

The recursive statistics (1/2)

- **Window regression** on the log price y_t, over the sample fractions $[r_1, r_2]$

$$\Delta y_t = a + \delta y_{t-1} + \sum_{j=1}^{k} \phi_j \Delta y_{t-j} + e_t, \qquad t = \lfloor r_1 T \rfloor, \ldots, \lfloor r_2 T \rfloor$$

  - $\Delta y_t = y_t - y_{t-1}$; a: intercept; δ: the coefficient tested; ϕ_j: coefficients of k lagged differences; e_t: error; T: sample size; $\lfloor \cdot \rfloor$: integer part
  - $\mathrm{ADF}_{r_1}^{r_2}$: the t-statistic of δ on this window
- H_0: $\delta = 0$ (unit root) against H_1: $\delta > 0$ (explosive): a **right-tailed** test
  - minimum window fraction $r_0 = 0.01 + 1.8/\sqrt{T}$
- **Suprema** over windows Phillips, Wu and Yu (2011), Phillips, Shi and Yu (2015a)

$$\mathrm{SADF} = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}, \quad \mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}, \quad \mathrm{GSADF} = \sup_{r_2} \mathrm{BSADF}_{r_2}$$

  - SADF: windows starting at the first observation; BSADF: all windows ending at r_2 (used to date); GSADF: all windows

The recursive statistics (2/2)

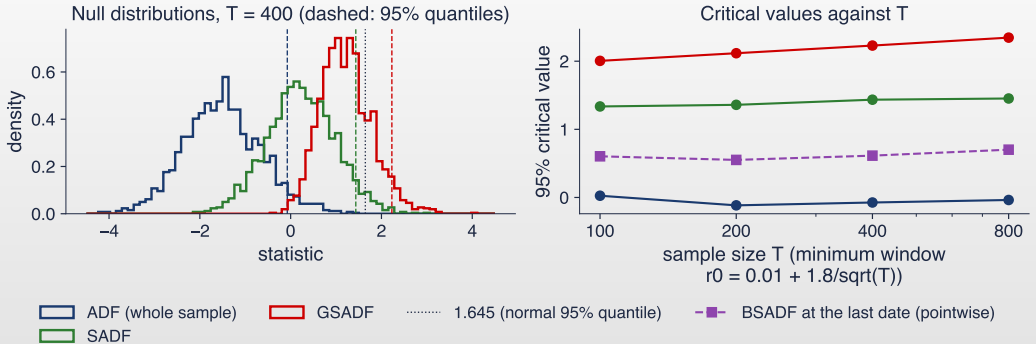
- Null limit (PSY, Theorem 1), with $r_w = r_2 - r_1$ and integrals over $[r_1, r_2]$:

$$\mathrm{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w \left[W(r_2)^2 - W(r_1)^2 - r_w\right] - \int W\, dr \, \left[W(r_2) - W(r_1)\right]}{r_w^{1/2} \left\{ r_w \int W^2 dr - \left(\int W\, dr\right)^2 \right\}^{1/2}}$$

  - W: a standard Brownian motion; $\Rightarrow$: convergence in distribution as $T \to \infty$
  - pivotal: free of σ and of the drift, so critical values depend only on r_0, the deterministic terms and T in finite samples

- Computation: for $T = 400$ there are 65 341 window regressions; cumulated cross-products give all of them in $O(T^2)$ operations

Null distributions and critical values



- Null model $y_t = T^{-1} + y_{t-1} + e_t$, $e_t \sim N(0,1)$, no lags; 95% critical values at $T = 100, 200, 400, 800$: SADF 1.34, 1.36, 1.43, 1.45; GSADF 2.01, 2.12, 2.23, 2.35

ATS_ch16_recursive_tests

Interpreting the null distributions

- The whole-sample ADF has a negative 95% quantile (-0.07 at $T = 400$): using 1.645 would almost never reject, the right tail of the Dickey–Fuller law is short
- Each supremum shifts the distribution to the right: SADF searches over end points, GSADF over start and end points; the GSADF bar is the price of the search
- The GSADF critical value rises with T (2.01 to 2.35) although r_0 falls: more windows, a larger maximum; tabulate critical values for your own T, r_0 and lag order
- The pointwise 95% quantile of BSADF at the last date (0.61 at $T = 400$) is far below the GSADF one: a single date is tested at 5%, the whole path is not

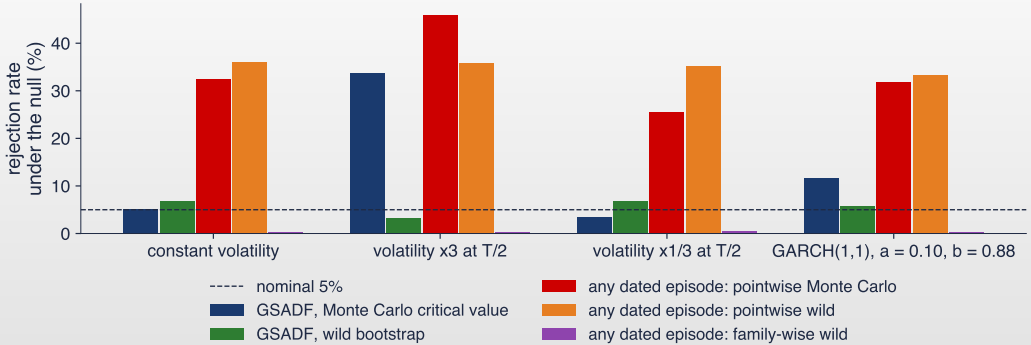
Changing volatility and the wild bootstrap

- The pivotal limit assumes homoskedastic shocks; with a volatility shift the limit involves the variance profile, and Monte Carlo critical values give the wrong size Harvey et al. (2016)
  - a late rise in volatility looks like acceleration; a fall in volatility makes the test conservative
- **Wild bootstrap** Phillips and Shi (2020): fit the null $\Delta y_t = \mu + \sum_j \phi_j \Delta y_{t-j} + e_t$, draw Rademacher signs $w_t = \pm 1$, build $\Delta y_t^* = \sum_j \hat{\phi}_j \Delta y_{t-j}^* + w_t \hat{e}_t$, cumulate, recompute the statistics
  - the bootstrap series has a unit root by construction and the **same volatility pattern** $|\hat{e}_t|$ as the data
  - critical values: pointwise quantiles of BSADF, the quantile of the maximum (GSADF), or the quantile of the maximum over a moving window
- Alternatives for non-stationary volatility: weighted and sign-based statistics Harvey, Leybourne and Zu (2020); end-of-sample tests Astill et al. (2017)

Date-stamping as a multiple-testing problem

- Date-stamping compares BSADF_{r_2} with a **pointwise** critical value at every date: each date has a 5% false-alarm chance, but a sample has hundreds of dates
  - BSADF is highly autocorrelated in r_2, so false alarms come in runs: the minimum-duration rule removes the shortest runs only
- **Family-wise error rate** (FWER): the probability of at least one false alarm over a set of dates
  - over the whole sample: compare BSADF with the 95% quantile of its maximum, i.e. the GSADF critical value
  - over a moving window of τ_b dates Phillips and Shi (2020): the 95% bootstrap quantile of $\max_{s\in(t-\tau_b,t]} \mathrm{BSADF}_s$; here $\tau_b = 52$ weeks
- Asymptotically, consistent dating needs critical values that diverge slowly, so that the false-alarm probability vanishes Phillips, Shi and Yu (2015b); in practice a fixed 5% at every date is used, and its cost must be measured

Size under changing volatility and across dates



- 400 null paths of $T = 200$ per design, each with its own wild bootstrap (199 draws); a dated episode needs 6 consecutive exceedances; nominal level 5%

ATS_ch16_size_wild_bootstrap

Interpreting the size study

- GSADF with Monte Carlo critical values: 5.2% with constant volatility, 33.8% after a late tripling, 11.8% under GARCH; the wild bootstrap gives 6.8%, 3.2% and 5.8%
- Pointwise dating finds at least one episode in 32.5% (Monte Carlo) and 36.0% (wild) of paths **without any bubble**, even with constant volatility
- With the family-wise threshold over the whole sample, false episodes fall to 0.2%–0.5%: below 5%, because an episode also needs 6 consecutive exceedances; the cost is power and later dates
- Lesson: report which error rate a dated episode controls; a 5% pointwise test on every week is not a 5% test of the sample

Recap: recursive tests

- SADF, GSADF and BSADF are suprema of window ADF statistics with a pivotal null limit; critical values must match T, r_0, the deterministic terms and the lags
- Changing volatility distorts the size; the wild bootstrap keeps the volatility pattern and restores it
- Pointwise dating has a high false-episode rate over a sample; family-wise thresholds control it at the cost of power

Date-stamping rules and their accuracy

- **Origination and termination** Phillips, Shi and Yu (2015a): the first date where BSADF exceeds its critical value, and the first later date where it falls back below

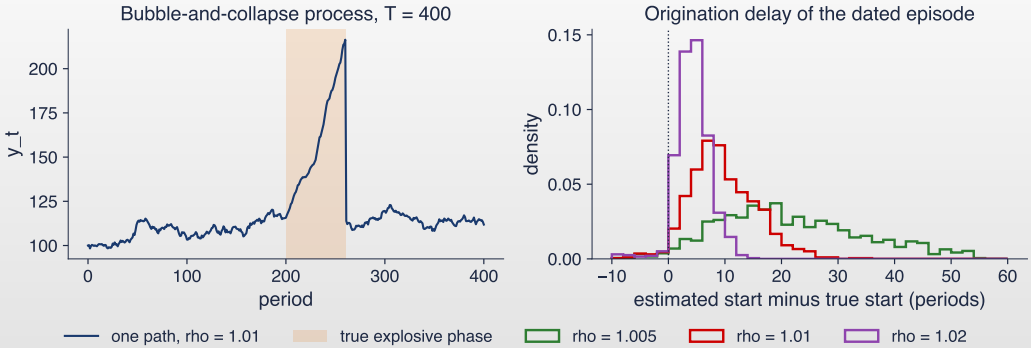
$$\hat{r}_e = \inf\{r_2 \geq r_0 : \mathrm{BSADF}_{r_2} > cv_{r_2}\}, \qquad \hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \mathrm{BSADF}_{r_2} < cv_{r_2}\}$$

  - cv_{r_2}: the pointwise 95% critical value at r_2; $\delta \ln(T)/T$: a minimum duration (with δ a tuning constant)
  - here: an episode is a run of at least $\lceil \ln T \rceil$ consecutive exceedances, dated from the first to the last one (as in `exuber` Vasilopoulos, Pavlidis and Martínez-García (2022))
- In real time an episode is **confirmed** only after the minimum duration: the alarm date is the start plus $\lceil \ln T \rceil - 1$
- Consistency of $(\hat{r}_e, \hat{r}_f)$ holds for mildly explosive bubbles of fixed fractional duration Phillips, Shi and Yu (2015b)
  - the start is dated late, because a window must first accumulate explosive observations
  - improved start and end estimators that use the whole bubble period: Harvey, Leybourne and Sollis (2017)

Date-stamping: design of the experiment

- Simulated paths: random walk from $y_0 = 100$, then an explosive phase, a collapse, and a random walk again
  - explosive phase $y_t = \rho y_{t-1} + e_t$, $\rho > 1$, on periods 200–260; e_t: Normal noise
  - collapse to the pre-bubble level at the end of the explosive phase
  - $T = 400$, 1 000 paths for each value of ρ
- Questions: how often is the bubble found, how late is its start dated, how often is a false episode dated before it?

Delay in dating the bubble start



- Delay of the first dated exceedance after the true start (periods); pointwise 95% Monte Carlo critical values; growth over the bubble ρ^{60}: 1.35, 1.82 and 3.28

ATS_ch16_date_stamping

Interpreting the date-stamping experiment

- Detection rates 94.9%, 100.0% and 100.0% for $\rho = 1.005, 1.01, 1.02$: a bubble of 60 periods is almost always found
- Median start delays 19, 8 and 4 periods (interquartile ranges 11–29, 5–13, 2–6); confirmation adds $6 - 1$ periods: medians 24, 13, 9
- For the slowest bubble the alarm comes after about a third of its life: a detector of **ongoing** exuberance, not of its birth
- In 26%–31% of paths a false episode is dated **before** the bubble starts: the multiplicity of the previous section, now inside a true-bubble design

Real-time monitoring (1/2)

- Monitoring: a training sample of n observations without a bubble, then a decision at each new date $t > n$; the error rate is over the **whole** monitoring horizon Chu, Stinchcombe and White (1996)
- **CUSUM** of returns: cumulated deviations of the new returns from the training mean, in units of the training volatility

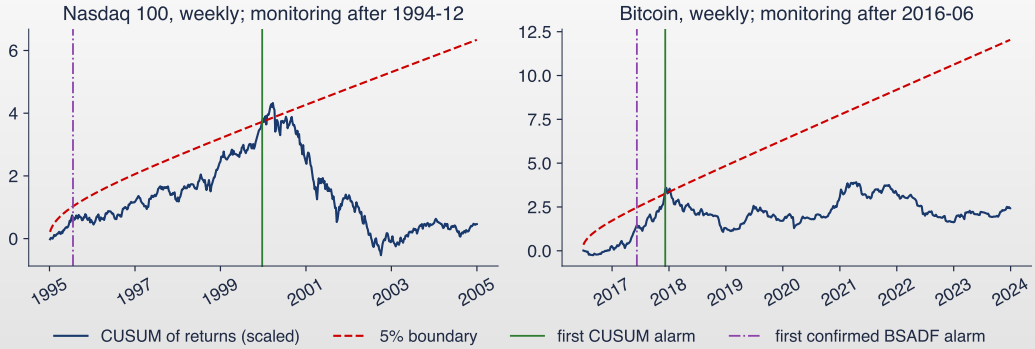
$$S_t = \sum_{j=n+1}^{t} \frac{r_j - \bar{r}_n}{\hat{\sigma}_n \sqrt{n}}, \qquad \text{alarm when } S_t > b_t = \sqrt{\tfrac{t-n}{n} \cdot \tfrac{t}{n} \left[a^2 + \ln \tfrac{t}{t-n} \right]}$$

  - r_j: return of period j; $\bar{r}_n$, $\hat{\sigma}_n$: mean and standard deviation of the training returns; b_t: the boundary, which widens with t; a: a constant that sets the error rate
- Error rate over the whole horizon (proof in the Appendix)
  - $\Pr\{S_t > b_t \text{ for some } t\} \to 1 - \Phi(a) + a\varphi(a)$; Φ, φ: standard Normal distribution and density functions
  - 5% gives $a = 2.50$, $a^2 = 6.25$

Real-time monitoring (2/2)

- Simulated size, $n = 100$, horizon $5n$, 2 000 paths
  - 3.5% with i.i.d. returns, 6.0% with GARCH returns
- Homm and Breitung (2012) compare bubble tests and monitoring schemes and recommend monitoring of this kind for real time
  - Astill et al. (2018) give monitoring statistics with controlled size for explosive bubbles
- BSADF in real time is also a monitor: BSADF_t uses only data up to t, but its pointwise critical values do not control the error over the horizon

Monitoring the Nasdaq 100 and Bitcoin



- Weekly log returns; training: 1990–1994 (Nasdaq 100, $n = 260$) and September 2014 – June 2016 (Bitcoin, $n = 92$); BSADF with pointwise Monte Carlo critical values

ATS_ch16_monitoring

Interpreting the monitoring charts

- Nasdaq 100: the CUSUM alarm comes on 24 December 1999, three months before the peak of 24 March 2000; the first confirmed BSADF alarm after training comes on 21 July 1995, more than four years earlier
- Bitcoin: CUSUM on 8 December 2017, at the top of the 2017 run-up; BSADF on 9 June 2017
  - the boundary widens with $\sqrt{t \ln t}$, so after 2018 the 2021 rally cannot reach it
- CUSUM detects a change in the mean return (persistent acceleration), not explosiveness as such; it is slow, but its size is controlled over the whole horizon
- BSADF reacts early and often: an early alarm that is followed by years of further gains is costly to anyone who acts on it

Evaluating early warnings

- Define the **event** before looking at alarms: here a fall of at least 20% within 182 days after the date; define the **alarm** rule and the evaluation dates (every fifth trading day)
- Hit rate $H = \Pr(\text{alarm} \mid \text{event})$, false-alarm rate $F = \Pr(\text{alarm} \mid \text{no event})$, base rate $\pi_0 = \Pr(\text{event})$
  - **precision** by Bayes: $\Pr(\text{event} \mid \text{alarm}) = H\pi_0 / [H\pi_0 + F(1-\pi_0)]$; with $\pi_0 = 5\%$, $H = 80\%$, $F = 20\%$ it is only 17%
  - the noise-to-signal ratio F/H of the early-warning literature Kaminsky, Lizondo and Reinhart (1998)
- **ROC curve**: (F, H) for every threshold of a continuous score; **AUC** $= \Pr(\text{score of an event date} > \text{score of a non-event date})$; 0.5 means no skill
- Overlapping horizons make evaluation dates strongly dependent: intervals for the AUC need a moving-block bootstrap (Chapter 0), and a few crashes carry the whole sample

Recap: date-stamping and early warning

- BSADF dates the start late (after a run of explosive observations) and confirms it later still; slow bubbles are found after a large part of their life
- Monitoring with a boundary controls the error over the whole horizon; pointwise BSADF does not
- Evaluate alarms with hit rate, false-alarm rate, precision at the base rate and ROC/AUC, with block-bootstrap intervals

Testing the ratio, the price and the fundamental

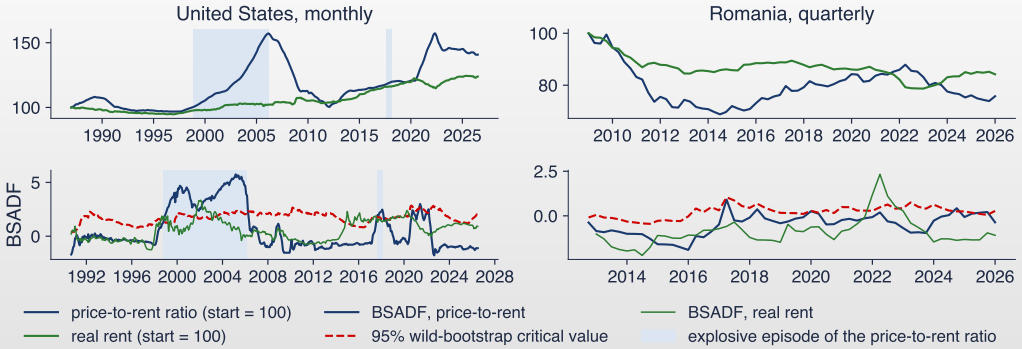


A foreclosed home in the United States, 2008

Photo: Brendel (2008); CC BY-SA 3.0; Wikimedia Commons

- Under the present-value model the price-to-rent (or price-dividend) ratio is stationary; an explosive **ratio** with a non-explosive **fundamental** points to a bubble
- If rents themselves explode, an explosive price is not evidence of a bubble: test both (the logic of Diba and Grossman (1988b) applied with PSY)
- Landmarks: Phillips and Yu (2011) date the US price-to-rent exuberance before the subprime crisis; Pavlidis et al. (2016) apply GSADF to the international housing panel of the Dallas Fed
- Monthly house prices are smoothed (repeat-sales indices, appraisals): their changes are autocorrelated, so the lag order k matters; here k by BIC, up to 6 months

US and Romanian housing: ratio and fundamental



⊡ Log price-to-rent and log real rent; BSADF with k lags (US ratio $k = 1$, rent $k = 1$) and pointwise 95% wild-bootstrap critical values (999 draws); shaded: dated episodes of the ratio

ATS_ch16_housing

Interpreting the housing tests

- US price-to-rent: GSADF 5.75 against the wild-bootstrap value 4.47; the main episode runs from November 1998 to March 2006, the boom before the subprime crisis found by Phillips and Yu (2011)
- US real rent: GSADF 3.35 against 3.25, with episodes February 1998–March 1999; June 2001–September 2004; September 2014–May 2021; September 2023–April 2026: the fundamental also accelerated, so the 2014–2022 price rise is partly a rent story
- US real house price alone: episodes February 2002–March 2006; December 2020–May 2022; the 2020–2022 surge is explosive in the price but not in the ratio, unlike 1998–2006
- Romania (quarterly, $T = 68$): GSADF of the ratio 0.91 against 1.38, of the real price 0.88 against 2.11: no explosive episode since 2009; the HICP rent index jumps by 33% in April 2026, so the sample ends in March 2026

Recap: fundamentals against prices

- Test the ratio and the fundamental separately: an explosive ratio with a calm fundamental is the bubble signature
- Smoothed housing indices need lag augmentation and a bootstrap under the AR null; without lags the tests reject almost everywhere
- Short samples (Romania since 2009) have little power: no rejection is not evidence of no bubble

The LPPLS model



Didier Sornette, 2012

Photo: Didier Sornette (2012); CC BY-SA 3.0 de; Wikimedia Commons

- Johansen, Ledoit and Sornette (2000): the log price grows faster than exponentially towards a critical time t_c, with oscillations that speed up

$$\ln p(t) = A + B(t_c - t)^m + C_1(t_c - t)^m \cos(\omega \ln(t_c - t)) + C_2(t_c - t)^m \sin(\omega \ln(t_c - t))$$

  - A: log price at t_c; $B < 0$ and $0 < m < 1$: super-exponential growth; C_1, C_2: amplitudes of the oscillations; ω: their log-frequency
- Two steps Filimonov and Sornette (2013): for given (t_c, m, ω) the four linear parameters are OLS; minimise the concentrated sum of squares over three nonlinear parameters
- Qualified fits: the filter of Shu and Zhu (2020) (bounds on m, ω, t_c, number of oscillations, damping, relative error, Lomb test, stationary residuals)
- **Confidence indicator**: the share of qualified fits over many windows ending at the same date (here 29 windows of 50–750 days)

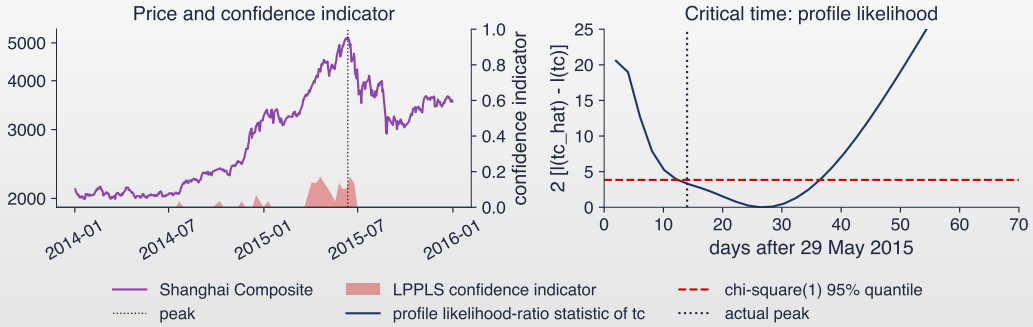
Inference on the critical time

- The point estimate $\hat{t}_c$ moves with the window and the data; an **interval** is needed
- **Profile likelihood**: for each t_c, the other parameters are set to their best values; with Gaussian errors

$$\ell(t_c) = -\frac{n}{2} \ln \left[\mathrm{SSR}(t_c)/n \right], \qquad 95\% \text{ interval} : \{ t_c : 2[\ell(\hat{t}_c) - \ell(t_c)] \leq \chi^2_{1;0.95} = 3.84 \}$$

  - n: number of days in the window; $\mathrm{SSR}(t_c)$: the smallest sum of squared residuals for this t_c
  - too narrow when residuals are autocorrelated or when nuisance parameters are many
- Filimonov, Demos and Sornette (2017): the modified profile likelihood (Barndorff-Nielsen) corrects for the nuisance parameters; intervals for t_c become wider and asymmetric
- Demos and Sornette (2017) ask whether the birth or the burst of a bubble is easier to diagnose: the start time is far better constrained by the data than t_c

Shanghai 2015: indicator and critical time



- Left: confidence indicator every fifth trading day; right: profile likelihood of t_c for the window from the low of 20 March 2014 to 29 May 2015 (293 days), as in the real-time analysis of Sornette et al. (2015)

ATS_ch16_lppls_inference

Interpreting the Shanghai fit

- The fit ending on 29 May 2015 passes the whole filter: $m = 0.41$, $\omega = 6.28$, 5.7 oscillations, damping 1.16; $\hat{t}_c =$ 24 June 2015
- Profile-likelihood interval: 12 June 2015 to 2 July 2015 (20 days); the peak of 12 June 2015 lies at its left edge: a good call that is fragile to the window
- The indicator is low: at most 0.17 (on 20 April 2015); few windows qualify at once, so the indicator flags a regime, not a date
- One success is an anecdote: the next section counts alarms over decades, with base rates

Spectral views of log-periodicity and the critique

- The oscillation $\cos(\omega \ln(t_c - t))$ is periodic in $\ln(t_c - t)$
  - test: the Lomb periodogram of the detrended residual against $\ln(t_c - t)$ Lomb (1976)
  - nonparametric alternative: the (H, q)-derivative analysis Zhou and Sornette (2003)
  - in calendar time its instantaneous frequency is $\omega/[2\pi(t_c - t)]$: oscillations speed up towards t_c; the Hilbert transform of the residual gives this phase directly (Chapter 11 for the tools)
- Critique
  - log-periodic peaks appear in noise after fitting and detrending Feigenbaum (2001)
  - out-of-sample crash calls perform poorly Brée and Joseph (2013); estimation is fragile Geraskin and Fantazzini (2013)
  - replies on the stochastic formulation, filters and multi-window indicators: Sornette et al. (2013)
- A fair comparison with PSY uses the same events, dates and error measures: next section

Recap: LPPLS

- LPPLS models super-exponential growth with log-periodic oscillations towards a critical time; the confidence indicator aggregates qualified fits over windows
- The critical time needs an interval: profile or modified profile likelihood; the birth of a bubble is easier to pin down than its burst
- Spectral tests of log-periodicity are easy to fool; judge LPPLS by its alarms, not by its fits

Five bubble episodes: data and dating procedure

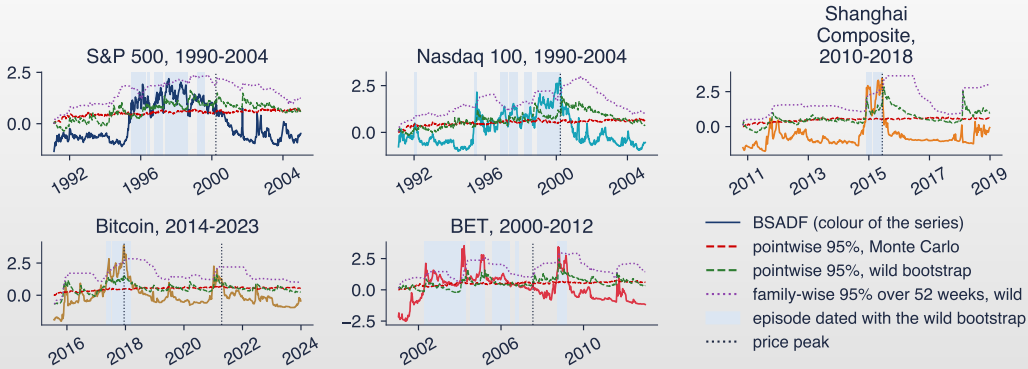


Shanghai Stock Exchange, Pudong, 2008

Photo: Baycrest (2008); CC BY-SA 2.5; Wikimedia Commons

- Weekly log prices (Friday close): S&P 500 and Nasdaq 100 1990–2004, Shanghai Composite 2010–2018, Bitcoin September 2014 – 2023, BET 2000–2012
- No lags; $r_0 = 0.01 + 1.8/\sqrt{T}$; minimum duration $\lceil \ln T \rceil$ weeks; Monte Carlo (1000 paths) and wild-bootstrap (499 draws) critical values
- Episodes dated with pointwise wild values and with the family-wise wild threshold over 52 weeks; the alarm date is the confirmation date
- Case studies: Phillips, Wu and Yu (2011) (Nasdaq), Phillips, Shi and Yu (2015a) (S&P 500), Sornette et al. (2015) (Shanghai), Gerlach, Demos and Sornette (2019) (Bitcoin); for the BET 2007 see also the further reading Pele and Mazurencu-Marinescu (2012)

BSADF of the five episodes



- Shaded: episodes dated with the pointwise wild bootstrap; dotted vertical lines: price peaks

ATS_ch16_episodes

Interpreting the five tests

Series	T	GSADF	cv MC	cv wild	episodes: MC / wild / FW	weeks: MC / wild
S&P 500	782	2.19	2.28	2.91	5 / 6 / 0	228 / 167
Nasdaq 100	782	2.92	2.28	3.22	6 / 6 / 1	210 / 159
Shanghai	460	3.46	2.15	3.90	1 / 2 / 1	31 / 27
Bitcoin	484	3.85	2.18	3.15	2 / 2 / 1	75 / 43
BET	663	3.53	2.29	3.21	3 / 7 / 1	213 / 217

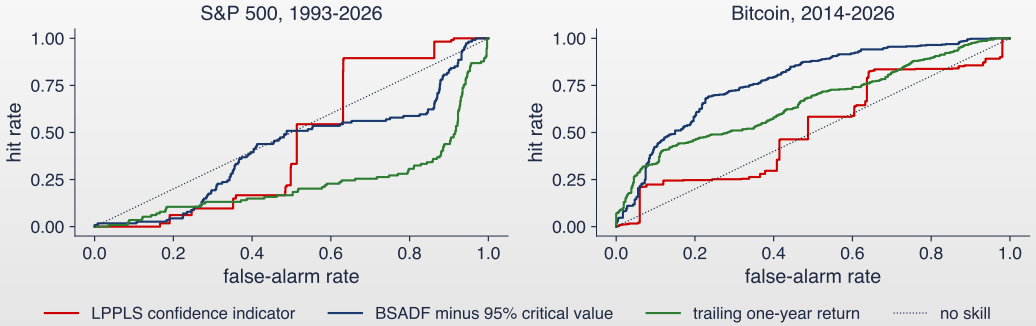
- S&P 500: GSADF below both critical values, yet pointwise dating marks 6 episodes: the multiplicity trap on real data; no family-wise episode
- The wild critical values exceed the Monte Carlo ones for every series (volatility clusters): Bitcoin and BET still reject; the Nasdaq (2.92 against 3.22) and Shanghai (3.46 against 3.90) do not
- MC: Monte Carlo; FW: family-wise wild threshold over 52 weeks; cv: 95% critical value of GSADF

Alarms before the peaks

Peak	first alarm (2 years before)	lead (weeks)	active at peak	FW lead	fall in 1 year
S&P 500, 24 March 2000	16 April 1999	49	no	–	25%
Nasdaq 100, 24 March 2000	24 April 1998	100	yes	13	65%
Shanghai, 12 June 2015	9 January 2015	22	yes	22	47%
Bitcoin, 15 December 2017	23 June 2017	25	yes	–	82%
Bitcoin, 16 April 2021	none	–	no	–	49%
BET, 20 July 2007	2 September 2005	98	no	–	46%

- Every major peak is preceded by a pointwise alarm except Bitcoin 2021; leads of half a year to two years: too early to time a crash, informative about the regime
- BET 2007: the alarm of 2 September 2005 is followed by almost two years of gains before the peak and a fall of 46%; family-wise control removes most alarms
- Counting only peaks hides the false alarms: alarms that were not followed by a crash appear only when every date is evaluated

Early-warning value over decades



- Event: a fall of 20% within 182 days; evaluation every fifth trading day (S&P 500 1 672 dates, Bitcoin 893); scores: LPPLS confidence indicator, BSADF minus its pointwise 95% critical value (weekly, real time), trailing one-year return

ATS_ch16_early_warning

Interpreting the early-warning evaluation

- Base rates: 6.8% of S&P 500 dates and 47.6% of Bitcoin dates are followed by a 20% fall: the same alarm means very different things in the two markets
- S&P 500, AUC with 90% block-bootstrap intervals: indicator 0.39 [0.32, 0.48], BSADF 0.41 [0.28, 0.58], momentum 0.23 [0.10, 0.40]: all below 0.5
- Most S&P 500 dates followed by a 20% fall lie inside bear markets (2000–2002, 2008), when exuberance scores are low, and few at the top of a run-up
  - so the scores are **lower** before the events; BSADF alarms have precision 1% against a base rate of 6.8%
- Bitcoin: BSADF has skill, AUC 0.77 [0.69, 0.85], above momentum 0.66; its alarms reach precision 79% against a base rate of 47.6% (hit rate 32%, false alarms 8%)
- The LPPLS indicator is positive on 35.9% of S&P 500 dates and 4.6% of Bitcoin dates; its AUC is 0.39 and 0.49: under this protocol it does not separate pre-crash dates from the rest

Recap: the applications

- Wild-bootstrap critical values are higher than Monte Carlo ones on real data; some textbook rejections do not survive
- Major peaks are usually preceded by alarms, but with long and variable leads; family-wise control removes most of them
- Over decades, precision must be read against the base rate
  - exuberance scores help for Bitcoin, where crashes follow run-ups
  - they fail for the S&P 500, where most pre-crash dates lie inside bear markets

An open question

- The question
  - do explosive-root and LPPLS alarms carry information about large falls beyond momentum and volatility?
  - once base rates, overlapping horizons and the search over assets are accounted for
- A testable form
  - formal: in a logit or probit of the event on lagged momentum, volatility and the alarm score, H_0: the alarm coefficient is zero, with block-bootstrap inference and a pre-registered panel of assets
  - falsified by a significant out-of-sample gain in a proper score (log score, Brier) over the momentum-volatility model, pooled across assets
- Why it matters
  - regulators and investors read bubble indicators as warnings
  - most evidence comes from a few famous episodes chosen after the fact
  - literature to start from: Phillips and Shi (2020), Astill et al. (2018), Demos and Sornette (2017), Brée and Joseph (2013), Homm and Breitung (2012)

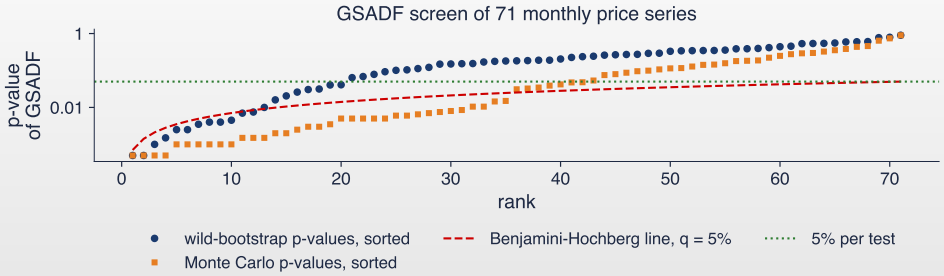
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2015 that evaluate bubble detectors out of sample with false-alarm rates; give DOIs.` Then check every DOI and the reported numbers
  - **hypothesis**: `Write a pre-registration: assets, sample, event definition, scores, baseline model, proper score, test and block length.`
  - **code and replication**: ask for a vectorised BSADF; reproduce a known number first (the GSADF of the Nasdaq 100 here, or PSY critical values)
  - **critique**: `Act as a hostile referee: list every way this evaluation could overstate the value of the alarms.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, the title matches, the result is in the paper)
- Right tail, not left: an AI-written ADF often uses left-tail critical values or the 1.645 normal quantile
- Critical values simulated for the same T, r_0, lags and deterministic terms; wild bootstrap when volatility changes
- No look-ahead: the minimum window, the lag order and the scaling are fixed from data available at each date
- Multiplicity over dates and over assets: which error rate does a reported bubble control?

Mini-case: multiple testing in a GSADF screen of many assets



- GSADF on the monthly log price of 71 series of the course data (from 1990 or the first month), p-values from 1999 wild-bootstrap and 1999 Monte Carlo draws (smallest attainable p-value: 0.0005)
- Rejections at 5%: 42 with Monte Carlo, 20 with the wild bootstrap (3.6 expected by chance); after Holm 2, after Benjamini–Hochberg 12. An AI summary such as "bubbles in 42 assets" is wrong twice: wrong null and no correction

ATS_ch16_ai_screen

Project idea

- **A pre-registered evaluation of bubble alarms on Central and Eastern European markets**
  - replicate: the date-stamping of Phillips, Shi and Yu (2015a) and the wild bootstrap with family-wise control of Phillips and Shi (2020) on BET, WIG20, BUX, PX and on housing price-to-rent ratios (Eurostat)
  - extend: compare BSADF, monitoring with a boundary and the LPPLS indicator as predictors of 20% falls, against momentum and volatility, with proper scores and block bootstrap
  - pre-register: assets, event definition, scores, horizons, tests and the decision rule, before running anything after 1 November 2026
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- A rational bubble is an explosive autoregressive component; it cannot be negative or start later, and periodic collapses hide it from whole-sample tests
- Explosive roots are learned at an exponential rate with Cauchy limits; mildly explosive asymptotics are invariant and justify the window tests
- Critical values must match the design; the wild bootstrap handles changing volatility; date-stamping needs family-wise control
- Dated starts are late, alarms are early relative to peaks, and precision depends on the base rate
- Test fundamentals as well as prices; judge LPPLS and PSY by the same alarms, events and scores

Key formulas

Object	Formula
Bubble	$P_t = F_t + B_t$, $\mathbb{E}_t B_{t+\mathbf{1}} = (1+r)B_t$
Fundamental, random-walk dividends	$F_t = D_t/r + \mu(1+r)/r^{\mathbf{2}}$
Explosive OLS limit	$\rho^n(\hat{\rho}-\rho)/(\rho^{\mathbf{2}}-1) \Rightarrow \mathcal{C}$
Recursive statistics	$\mathrm{BSADF}_{r_{\mathbf{2}}} = \sup_{r_{\mathbf{1}}} \mathrm{ADF}_{r_{\mathbf{1}}}^{r_{\mathbf{2}}}$, $\mathrm{GSADF} = \sup_{r_{\mathbf{2}}} \mathrm{BSADF}_{r_{\mathbf{2}}}$
Minimum window	$r_{\mathbf{0}} = 0.01 + 1.8/\sqrt{T}$
Monitoring boundary	$b_t = \sqrt{\frac{t-n}{n}\frac{t}{n}[a^{\mathbf{2}} + \ln\frac{t}{t-n}]}$, $1-\Phi(a)+a\varphi(a)=\alpha$
Precision of an alarm	$H\pi_{\mathbf{0}}/[H\pi_{\mathbf{0}} + F(1-\pi_{\mathbf{0}})]$
LPPLS	$\ln p = A + Bf + C_{\mathbf{1}} f\cos(\omega\ln(t_c-t)) + C_{\mathbf{2}} f\sin(\omega\ln(t_c-t))$, $f=(t_c-t)^m$

Self-assessment (1/2)

- Derive F_t when dividends follow $D_t = \mu + D_{t-1} + \varepsilon_t$; what is F_t for $r = 0.04$, $\mu = 0.1$, $D_t = 3$?
- Why can a rational bubble not start after the first trading day?
- Why does the whole-sample ADF fail against an Evans bubble?
- Why is the limit of $\hat{\rho}$ for a fixed explosive root Cauchy only with Gaussian errors?
- A sample has $T = 400$ weeks. What are r_0 and the minimum window?
- Why is the GSADF critical value larger than the SADF one?

Self-assessment (2/2)

- What does the wild bootstrap keep from the data, and what does it remove?
- An alarm has $H = 0.7$ and $F = 0.1$; the base rate is 4%. What is its precision?
- Why must the alarm date be the confirmation date and not the dated start?
- The price-to-rent ratio is explosive and real rents are explosive too. What do you conclude?
- Why is a profile-likelihood interval for t_c too narrow with autocorrelated residuals?
- A screen of 80 assets finds 20 GSADF rejections at 5%. Which two checks come first?

Self-assessment: answers (1/2)

- $F_t = D_t/r + \mu(1+r)/r^2 = 75 + 65 = 140$
- If $B_t = 0$, then $\mathbb{E}_t B_{t+1} = 0$ and $B_{t+1} \geq 0$ (no negative bubbles), so $B_{t+1} = 0$ almost surely; by induction the bubble stays at zero
- Collapses pair the largest falls with the highest levels and pull the OLS slope down; over the whole sample the series looks I(1) or stationary
- $\rho^n(\hat{\rho} - \rho)/(\rho^2 - 1) \approx X_n/Y_n$ with X_n, Y_n dominated by a few shocks: they are normal only if the shocks are normal; no CLT acts inside them
- $r_0 = 0.01 + 1.8/20 = 0.1$; minimum window $\lfloor 0.1 \cdot 400 \rfloor = 40$ weeks
- GSADF takes a supremum over more windows (all starts and ends), so under the null its maximum is larger

Self-assessment: answers (2/2)

- It keeps the volatility pattern $|\hat{e}_t|$ (and the short-run dynamics of the null model); random signs remove any explosive or predictable component
- $0.7 \cdot 0.04/(0.7 \cdot 0.04 + 0.1 \cdot 0.96) = 0.028/0.124 \approx 0.23$
- The start is known only after the minimum number of exceedances; using it as the alarm date would use future information
- No evidence of a bubble from these tests: the fundamental explains the acceleration; one would test whether the ratio explodes beyond what rents imply
- Autocorrelated residuals carry less information than n independent errors, and nuisance parameters are ignored: the likelihood is too sharp
- Whether the critical values allow for changing volatility (wild bootstrap); whether the number of rejections survives a multiple-testing correction (4 are expected by chance)

Further reading and links



The Bucharest Stock Exchange Palace, March 1928

Photo: Nicolae Ionescu (1928); public domain; Wikimedia Commons

- Surveys
  - Gürkaynak (2008); Shiller (2015); Kindleberger and Aliber (2005)
  - bubbles and later returns: Greenwood, Shleifer and You (2019)
- Implementation: the R package `exuber` Vasilopoulos, Pavlidis and Martínez-García (2022)
  - our Quantlets reproduce its statistics in Python
- Applications to markets, risk and policy: MFM, Chapter 17
- Further reading on the Bucharest Stock Exchange: Pele and Mazurencu-Marinescu (2012)
- The course ends here; the project defence is described in Chapter 15

Appendix: the mildly explosive Cauchy limit

- $\rho_n = 1 + c/k_n$; $X_n = k_n^{-1/2} \sum_{j=1}^{n} \rho_n^{-(n-j)-1} u_j$, $Y_n = k_n^{-1/2} \sum_{j=1}^{n} \rho_n^{-j} u_j$; weights $\rho_n^{-j} \approx e^{-cj/k_n}$ decay over $O(k_n)$ terms
- Lindeberg: no single weight dominates, so $(X_n, Y_n) \Rightarrow (X, Y)$, independent $N(0, \sigma^2/2c)$ ($\sum_j \rho_n^{-2j} \approx k_n/2c$); independence because X_n and Y_n load on the two ends of the sample and $k_n = o(n)$
- $(k_n \rho_n^n)^{-2} \sum_t y_{t-1}^2 \Rightarrow Y^2/2c$ and $(k_n \rho_n^n)^{-1} \sum_t y_{t-1} u_t \Rightarrow XY$, so $\frac{k_n \rho_n^n}{2c}(\hat{\rho} - \rho_n) \Rightarrow X/Y \sim \mathcal{C}$ Phillips and Magdalinos (2007)
- A ratio of two independent centred normals with equal variances is standard Cauchy: $X/Y = \tan\Theta$ with Θ uniform on $(-\pi/2, \pi/2)$

Back

Appendix: the monitoring boundary

- Under the null, $S_t \approx n^{-1/2}\sigma^{-1}[\sum_{j\le t}\varepsilon_j - (t/n)\sum_{j\le n}\varepsilon_j]$; with $s=(t-n)/n$: $S \Rightarrow Z(s) = W(1+s) - (1+s)W(1)$, $\mathrm{Var}Z(s) = s(1+s)$
- Time inversion: $Z(s) = s\,B(1+1/s)$ for a standard Brownian motion B; with $u = 1/s$ the event $Z(s) > \sqrt{s(1+s)[a^2 + \ln((1+s)/s)]}$ becomes $B(1+u) > \sqrt{(1+u)[a^2+\ln(1+u)]}$
- Robbins–Siegmund: $\Pr\{B(1+u) > \sqrt{(1+u)[a^2+\ln(1+u)]} \text{ for some } u \ge 0\} = 1 - \Phi(a) + a\varphi(a)$; the two-sided version doubles it, the form used by Chu, Stinchcombe and White (1996)
- In t: $s(1+s) = \frac{t-n}{n}\cdot\frac{t}{n}$ and $(1+s)/s = t/(t-n)$, which gives b_t; a finite horizon makes the test conservative

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